higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING
NQF LEVEL 2

(12050652)

1 March 2019 (X-Paper)
09:00–12:00

Calculators and drawing instruments may be used.

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Rule off across the page on completion of each question.
6. Drawings and sketches must be clear enough to show detail.
7. ALL drawings must be fully dimensioned and labelled.
8. Write neatly and legibly.

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QUESTION 1: INTRODUCTION TO CIVIL AND STRUCTURAL STEEL WORK

1.1 Portal frames were developed during World War II and are now the most common form of enclosure for spans of 20 m to 60 m.

1.1.1 What are *portal frames*? (2)

1.1.2 Give the purpose of portal-frame shapes. (2)

1.2 Define *corrosion* and name TWO corrosive agents. (3)

1.3 Define the term *legend* as it refers to drawing and give its purpose. (2)

1.4 Restrained beams and columns develop cracks over time.

FIGURE 1 shows a 600 mm deep beam restrained at both ends with knee connections.

Draw the knee section of the beam to a relevant scale and detail how it can be reinforced.

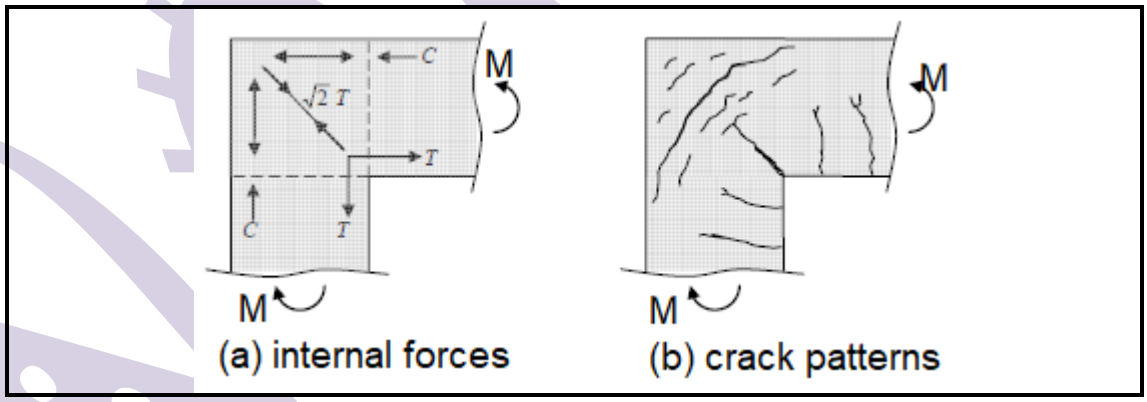


FIGURE 1 (15)

1.5 Differentiate between *mild steel* and *high-tensile steel*. (2 + 2) (4)

1.6 Describe the purpose of a retaining wall. (2)

1.7 State TWO things that the client/owner needs before instructing an architect to design a building. (2)

1.8 Differentiate between the *main contractor* and the *subcontractor*. (3)

1.9 A title panel is essential on a drawing to facilitate reference when prints are filed or folded and should be placed in the bottom right corner of the drawing sheet.

Name FIVE pieces of information that should be included in the title panel. (5)

[40]

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QUESTION 2: STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

2.1 Explain the function of TWO ingredients used to form concrete. (2 × 2) (4)

2.2 List THREE desired properties of concrete. (3)

2.3 Prepare a bar-bending schedule for the beam in FIGURE 2. The concrete cover = 40 mm.

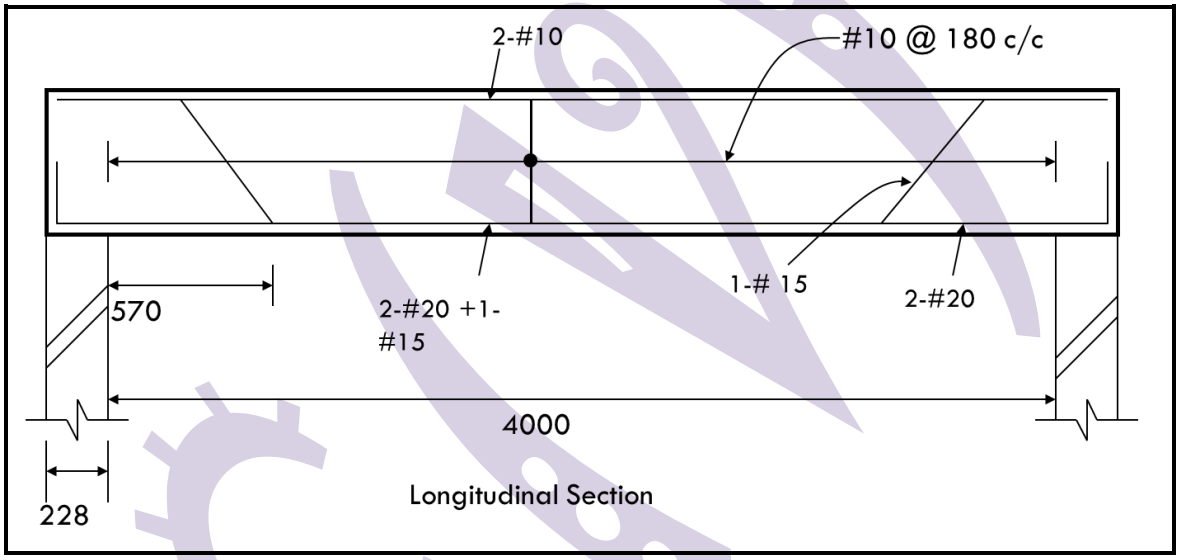


FIGURE 2 (15)

2.4 Complete the following paragraph by using the words in the list below. Write only the missing word next to the question number (2.4.1–2.4.6) in the ANSWER BOOK.

deterioration; durability; chemical; rain; weathering; wind

(2.4.1) … is a term used to describe the ability of a material to resist (2.4.2) … due to external forces such as (2.4.3) … and (2.4.4) … attack. Concrete is immune to the effects of (2.4.5) … and (2.4.6) … as long as proper precautions are taken. (6 × 1) (6)

2.5 Concrete should be tested to determine its compressive strength.

Rearrange the following steps of a cube test in chronological order:

- Fill at least six moulds with concrete.
- Pour a 50 mm layer of concrete into a mould and temper the concrete by using temping rods with exactly 45 blows.
- Use a hand trowel to level the concrete at the top of the mould.
- Cover the mould with a damp cloth.
- Repeat this step three times until the mould is completely filled.
- Label the mould with the date, the mixer's name and number of the mould. (6)

2.6 Describe the relationship between *stress* and *strain*. (2)

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2.7 Describe each of the following properties of steel:

2.7.1 Strength

2.7.2 Ductility

(2 × 2) (4)
[40]

QUESTION 3: BOLTS IN STRUCTURAL STEEL WORK

3.1 Give FOUR advantages of bolted connections. (4)

3.2 Explain what HSFG bolts are used for. (2)

3.3 Describe how holding-down bolts are fabricated. (2)

3.4 Give an outline of the specifications that govern the representation of ordinary bolts and HSFG bolt holes shown in fabrication drawings. (2)
[10]

QUESTION 4: WELDING SYMBOLS AND JOINTS

4.1 Various options are given as possible answers to the following question. Choose the answer and write only the letter (A–E) next to the question number (4.1) in the ANSWER BOOK. (1)

Which ONE of the following is NOT considered a type of joint:

A Butt
B T
C Fillet
D Corner
E Edge

4.2 Describe a *lap joint*. (2)

4.3 Explain each of the following regarding welding symbols:

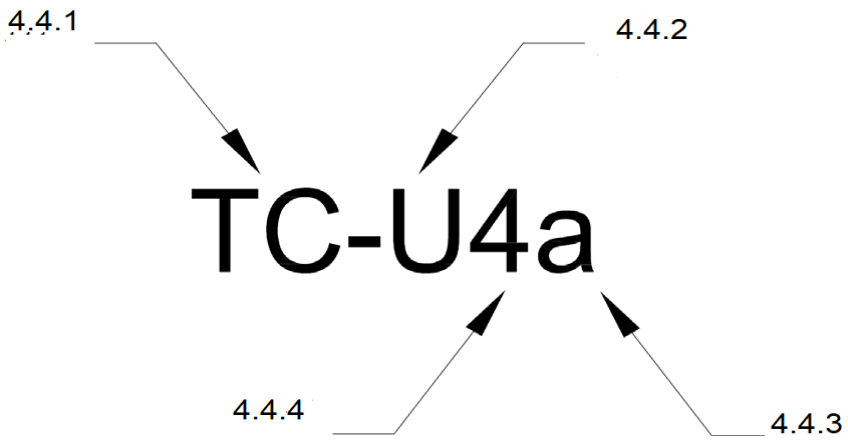
4.3.1 Reference line

4.3.2 Welding symbol

4.3.3 Weld length

(3 × 1) (3)

4.4 The picture below shows a weld designation.



Explain the respective components by writing only the answer next to the question number (4.4.1–4.4.4) in the ANSWER BOOK.

(4 × 1) (4)
[10]

TOTAL: 100

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